BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: KT-XII-21-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Code B

Note: Q.1 Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. No marks will be awarded for cutting, erasing or over writing.

Qi. The carbon of carbonyl group is , ...

(A) Sp^3 Hybridized
(B) Pyramidal in geometry
(C) Sp^2 Hybridized
(D) None of these

Qii. Hemoglobin is an example of.....

(A) Primary protein
(b) Secondary protein
(c) Tertiary protein
(d) Quaternary protein

Qiii. Petrochemicals are used in the manufacture of.....

(A) Polythene
(B) Detergents
(C) Fibers
(D) All of these

Qiv. 1,3 butadiene show absorbance at a wave length of .....nm.

(A) 217
(B) 171
(C) 400
(D) 435

Qv. Which one of the following is meta directing substance?

(A) –OH
(B) $–NH_2$
(C) –COOH
(D) $–C_2H_5$

Qvi. At $1720cm^{-1}$, an absorption peak is observed for ..... in IR spectrum.

(A) Aldehyde/ketone
(B) Ether
(C) Benzene
(D) None of these

Qvii. Which one of the following is acidic in nature?

(A) Al_2O_3 (B) CO_2

(C) CO (D) CaO

Qviii. The term 'en' in $[CO(en)_2Cl_2]$ is a ligand having ..... charge.

(A) Positive (B) Negative

(C) No (D) Both A & B

Qix. $TiCl_4$, a component of Zeigler-Natta catalyst is used in ..... of ethene.

(A) Polymerization (B) Hydration

(C) Reduction (D) Cracking

Qx. All of the following fractions are obtained by destructive distillation of coal except.....

(A) Coal tar (B) Refinery gas

(C) Coal gas (D) Coke

Qxi. The final stage of conversion of decaying plants into coal is.....

(A) Anthracite (B) Lignite

(C) Peat (D) Bituminous

Qxii. Magnesium burns in steam with a white flame producing .....

(A) $Mg(OH)_2+H_2O$ (B) $MgO+H_2$

(C) $Mg(OH)_2+H_2$ (D) $MgO+H_2O$

Qxiii. When acetylene is treated with HBr, we get.....

(A) Methyl bromide (B) Methyl chloride

(C) Ethyl bromide (D) Ethylene bromide

Qxiv. Which one of the following is synthetic polymer?

(A) Cellulose (B) Protein

(C) Nucleic acid (D) Polythene

Qxv. Which of the following represent the formula of an aromatic carboxylic acid?

(A) HCOOH (B) C_6H_5OH

(C) C_6H_5 COOH (D) CH_3CH_2OH

Qxvi. Which of the following are most reactive?

(A) Alkane (B) Alkene

(C) Alkyne (D) Benzene

Qxvii. Ketones are less reactive then aldehydes because of.....

(A) Resonance effect
(B) Steric effect
(C) Electronic effect
(D) Both B & C

Qxviii. Esterification is the reaction between .....

(A) An acid and ester
(B) An acid and alcohol
(C) An acid halide & ester
(D) An ester & sodium hydroxide

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

QQ.2. Attempt any TEN parts. All parts carry equal marks.

i. Explain how PAN is formed in our atmosphere through chemical reaction.

ii. Why Chloroacetic acid is stronger acid Acetic acid?

iii. Explain the reaction of Aldehyde with Hydrazine.

iv. Explain Carboxylic acids have high boiling points than corresponding alcohols.

v. Explain the Hydrolysis of Alkyle Nitriles.

vi. How will you differentiate E^1 from E^2 (in two points only for each)?

vii. Explain the geometry of coordination compound briefly.

viii. Discuss the Fischer-Tropsch process for the conversion of coal into petroleum.

ix. Write a short note on catalytic properties of transition elements.

x. How RX are prepared from Alcohols.

xi. Define and explain the term transition element.

xii. What is an Alloy? Give some examples.

xiii. What are Friedel-Craft's Alkylation and Friedel-Craft's Acylation reactions?

Section "C"

Maximum Marks: 27

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ.3.

a. How and why $BeCl_2$ is covalent? Explain through Fajan Rule.

b. Explain the structure and stability of chlorides of 4th A referring towards inert pair effect.

QQ.4.

a. Write down the chemical reaction of ethene with the following: (i) H_2 (ii) O_3 (iii) H_2O (iv) HCl

b. Describe any two methods of preparation of alkynes.

QQ.5.

a. What are phenols? Give methods of preparation of phenol with proper equation.

b. Explain the reaction of group II-A elements with oxygen.

QQ.6.

a. Discuss the method for preparation of diethyl ether from alcohol and sodium methoxide.

b. How and why modern method of analysis is superior over classical method of analysis?